Open-air JEt Mother/daughter (OJEM) Launch System

Application to ballistic missile defense using drone deployed nets

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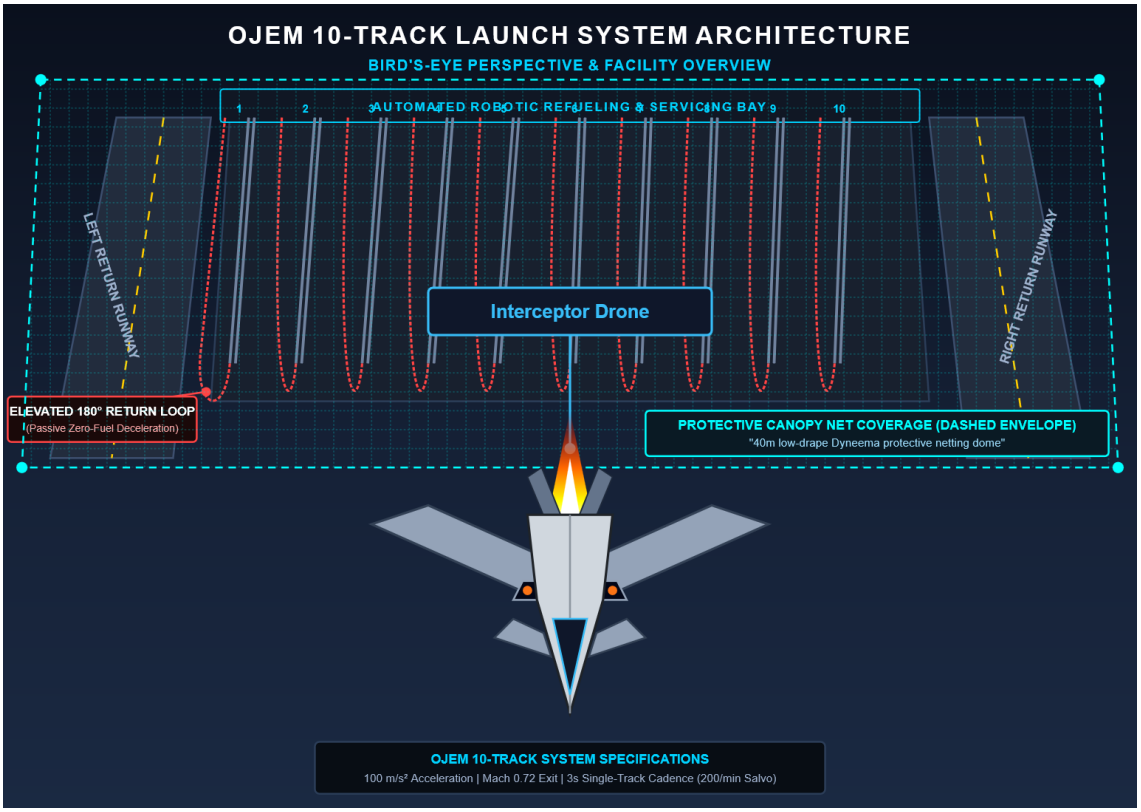
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This document and the technical specifications contained herein describe the Open-air JEt Mother/daughter (OJEM) Interceptor Launch System, which is the original work and invention of Ronald H. Smith.

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1. System Concept

The Open-air JEt Mother/daughter (OJEM) Launch System is a high-cadence, dual-role theater defense and strike architecture designed to launch high-velocity interceptor drones with a very favorable cost exchange ratio.

The system separates the launch vehicle into two distinct stages:

The Mother Sled: A reusable, 165 kg dry mass jet-propelled ground sled equipped with permanent magnet (PM) Halbach arrays for passive magnetic levitation and aerodynamic guide-arm airfoils. It accelerates along a 300-meter open-air steel rail pair, providing the initial kinetic energy.

The Daughter Interceptor: A 120 kg dry mass jet drone powered by a Pulse Detonation Engine (PDE) and deployed in a 50/50 warhead payload split:

Net-Carrier Interceptors (50% / 900 Units): Armed with high-tensile Dyneema barrier nets for non-explosive kinetic entanglement, wing shearing, and area denial.

Kinetic-Warhead Interceptors (50% / 900 Units): Armed with directional blast-fragmentation or direct-impact kinetic warheads for immediate hard-kill structural destruction.

Adding 50 kg fuel and 65 kg payload brings the total interceptor mass to 235 kg.

At the end of the 300-meter rail, the daughter drone ignites its PDE and releases into free flight, while the mother sled remains captive to the track, decelerates passively through an elevated 180° banked turn, and returns along parallel sidings for automated robotic refueling and re-coupling.

2. Launch Physics & Energy Expenditure

2.1 Acceleration and Kinematics

The launch profile specifies a constant horizontal acceleration $a = 100$ m/s^2 (≈ 10.2 g) over a track length $s = 300$ m. The combined initial launch mass is $M_0 = m_m + m_d = 235$ kg $+ 235$ kg $= 470$ kg.

Exit Velocity ($\boldsymbol{v_e}$)

$$v_e = \sqrt{2as} = \sqrt{2 \cdot (100 \text{ m/s}^2) \cdot (300 \text{ m})} = \sqrt{60{,}000} \text{ m/s} \approx 244.95 \text{ m/s} \quad (\approx \text{Mach } 0.72)$$

Launch Transit Time ($\boldsymbol{t_l}$)

$$t_l = \frac{v_e}{a} = \frac{244.95 \text{ m/s}}{100 \text{ m/s}^2} \approx 2.45 \text{ seconds}$$

2.2 Force Balance and Fuel Expenditure

The instantaneous thrust required from the mother sled must overcome system inertia and aerodynamic drag:

$$F_{thrust}(t) = M(t) \cdot a + F_{drag}(t)$$

Aerodynamic drag is calculated using $F_{drag}(t) = \frac{1}{2}\rho v(t)^2 C_D A$, where sea-level air density $\rho = 1.225$ kg/m^3, frontal area $A = 0.45$ m^2, and drag coefficient $C_D = 0.22$:

Peak Aerodynamic Drag ($v = 245$ m/s):

$$F_{drag,peak} = \frac{1}{2}(1.225)(245)^2(0.22)(0.45) \approx 3{,}638 \text{ N} \quad (3.64 \text{ kN})$$

Average Aerodynamic Drag ($F_{drag,avg}$):

$$F_{drag,avg} = \frac{1}{3}F_{drag,peak} \approx 1{,}213 \text{ N} \quad (1.21 \text{ kN})$$

Inertial Force ($F_{inertial}$):

$$F_{inertial} = 470 \text{ kg} \cdot 100 \text{ m/s}^2 = 47{,}000 \text{ N} \quad (47.0 \text{ kN})$$

Total Average Launch Thrust (F_{avg}):

$$F_{avg} \approx 47.0 \text{ kN} + 1.21 \text{ kN} = 48.21 \text{ kN}$$

Launch Fuel Expenditure ($\boldsymbol{m_{f,launch}}$)

Assuming jet booster propulsion with a specific impulse $I_{sp} = 220$ s:

Mass Flow Rate ($\dot{m}$):

$$\dot{m} = \frac{F_{avg}}{g_0 \cdot I_{sp}} = \frac{48{,}210 \text{ N}}{9.81 \text{ m/s}^2 \cdot 220 \text{ s}} \approx 22.34 \text{ kg/s}$$

Launch Leg Fuel Mass:

$$m_{f,launch} = \dot{m} \cdot t_l = 22.34 \text{ kg/s} \cdot 2.45 \text{ s} \approx 54.73 \text{ kg}$$

2.3 Return-Leg Dynamics and Zero-Fuel Return

Upon daughter release, the unladen and fuel depleted 165 kg mother sled enters an elevated 180° banked turn (10m vertical rise) and deploys variable-geometry speed brakes ($C_D > 1.2$), shedding 50% of its velocity down to 122.5 m/s. The sled coasting mechanics utilize gravity ascent, aerodynamic braking, and a terminal pneumatic/solid retrofire pulse to complete the return transit in 10.0 seconds without expending primary liquid jet propellant ($m_{f,return} = 0$ kg).

Total Fuel Consumed per Launch Cycle: 54.73 kg (Jet-A / propellant blend).

3. Physical Plant Architecture & Defensive Netting

A standard installation consists of a 10-track parallel array spanning an operational footprint of 500 m $\times$ 300 m.

3.1 Guideway and Return Geometry

Parallel Rails: 10 open-air steel guidance track pairs spaced 20 m apart (180 m total array width).

Parallel Return Forks: To prevent bottlenecking at the terminal deceleration area, each rail forks into 5 parallel return sidings leading to automated robotic servicing pits.

Flank Runways: Two 600 m asphalt runways border the left and right flanks of the complex to handle conventionally returning daughter drones.

3.2 Overhead Defensive Netting Canopy

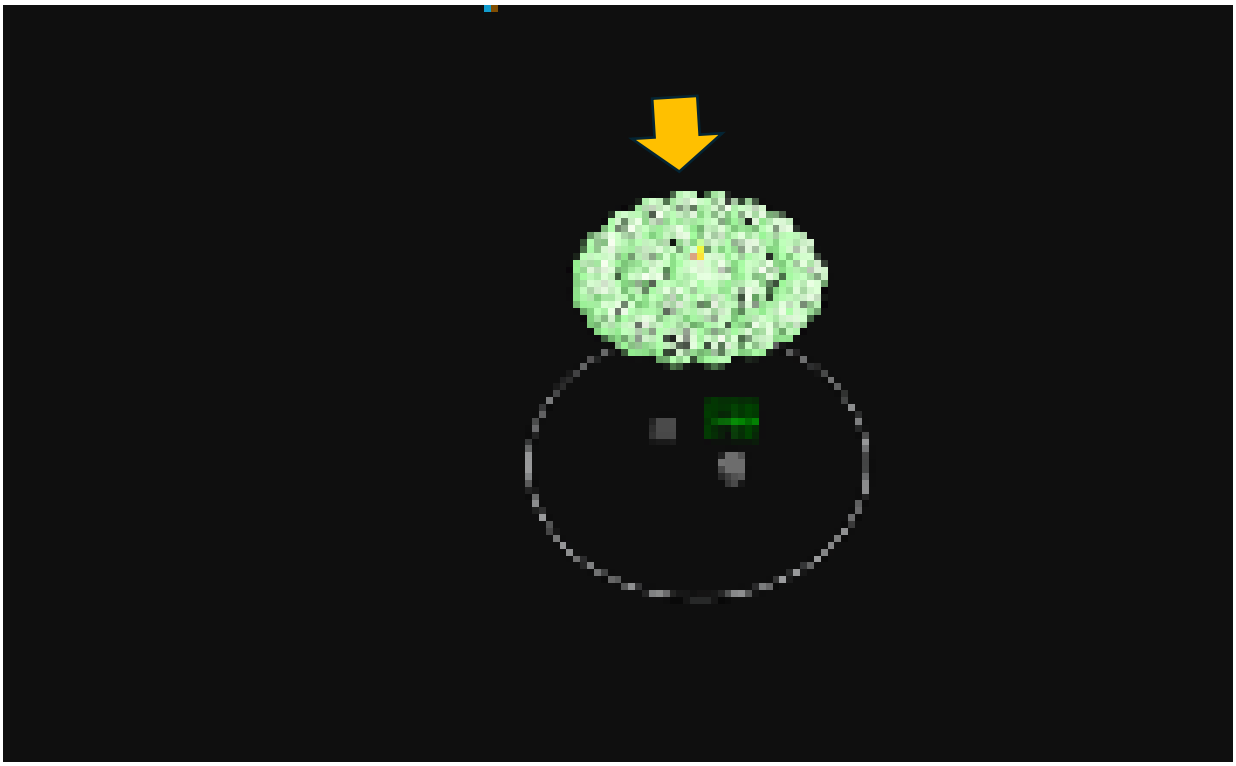
The entire complex is protected by a passive, high-tensile Dyneema SK78 netting canopy:

Suspension Masts: Eight 200m-high tapered composite/steel guyed masts support the primary overhead web.

Low-Drape Envelope: Across non-launch sectors (maintenance hangars, automated fuel depots, and taxi sidings), the net drapes down to a minimum height of 20 meters above ground level, forming a catenary protective dome that neutralizes incoming artillery shells, mortar rounds, and drone fragments.

3.3 Interceptor Web Payload Details

The interceptor can carry a deployable defensive web packaged in a canister. After release, the web spins open, while canted fins around its perimeter sustain its rotation and diameter. Its terminal descent rate is approximately 10 meters per second, providing about 100 seconds of dwell time. The figure below illustrates an array of thirty 200-meter-diameter webs protecting ground assets from incoming ballistic missiles.



3.3.1 Autorotation, Deployment, and Dwell

Canted perimeter fins convert the falling mesh into an autorotating, self-stabilizing rotor disc. Rather than allowing drag to draw the perimeter inward, the fins extract torque from the upward relative airflow during descent. The resulting angular velocity keeps the

perimeter weights under centrifugal tension, holding the 200 m-diameter, 31,415 m^2 web flat and taut without rigid spars.

Deployment sequence

Initial radial ejection only separates the lines and clears the central canister. As the perimeter weights move outward, the canted fins generate continuous torque and increasing angular velocity. Centrifugal acceleration, $a_c = \omega^2 r$, then progressively unfurls the web and maintains line tension throughout opening, replacing a violent snap-open event with a controlled three-second bloom.

Extending deployment to 3.0 seconds produces an average radial speed at a 100 m radius of about 33 m/s. Because kinetic energy scales with velocity squared, maintaining this modest deployment speed lowers line-breakage risk. It also permits low mass gas-generator squibs or low-impulse charges instead of high-velocity thrusters, reducing canister deadweight by an estimated 15–25 kg.

Descent and dwell

In autorotation, the web behaves aerodynamically like a low-disk-loading parachute, with an assumed drag coefficient of approximately 1.1–1.4. For a 150–200 kg system distributed across the 200 m disc, the predicted equilibrium descent rate is about 10 m/s.

During a three-second bloom, the system loses only about 30 m of altitude. A release at 2,000 m AGL therefore produces a fully deployed web near 1,970 m AGL and, under the stated descent-rate assumption, roughly 147 seconds before the web falls below 500 m. Releases between 1.5 and 2.0 km AGL provide approximately 100–150 seconds of useful dwell in the terminal dive corridor.

3.3.2 Web Design and Engagement Mechanism

The proposed web uses 0.8–1.0 mm Dyneema SK99 filament, with 0.9 mm as the nominal design point, arranged on a 0.70 m × 0.70 m pitch. The resulting 0.99 m diagonal yields approximately 83.7 km of cord, a bare-net mass of about 42–45 kg, and an estimated total canister assembly mass of 60–65 kg.

Geometric engagement

The intended defeat mechanism is aerodynamic destabilization rather than mechanical capture. A missile body approximately 0.92 m in diameter cannot pass cleanly through a 0.70 m square aperture, and a tailfin span of about 1.50 m would intersect multiple orthogonal mesh lines. This geometry is expected to engage several cords across the nose, fuselage, and control surfaces.

Load sharing and destabilization

At approximately 1,500 m/s, multiple lines would engage nearly simultaneously, distributing the transient load across an estimated cluster of 6–10 cords. A 1.0 mm SK99 cord is stated to have a breaking strength of about 1.6 kN. The design premise is that the combined, asymmetric impulse applied before cord failure produces a 1.5°–3° angular disturbance, generating aerodynamic bending moments sufficient to destabilize or structurally fail the vehicle.

3.3.3 Operational Employment and Engagement Window

The extended dwell period allows an array to be deployed within a broader timing window, permits one web to remain in a threat lane for staggered arrivals, and enables successive array launches to maintain overlapping coverage over protected assets.

Illustrative ballistic-intercept sequence

1. **T – 90 s:** Early-warning radar detects a midcourse ballistic trajectory toward the defended area.
2. **T – 60 s:** The sled begins its four-second stroke on the 300 m rail.
3. **T – 56 s:** The interceptor leaves the rail at 100 m/s and climbs into the terminal dive corridor.
4. **T – 34 s:** The interceptor reaches 2,000 m AGL.
5. **T – 31 s:** Sabots separate; the 200 m autorotating web is fully deployed and stabilized.
6. **T:** The incoming vehicle intersects the Dyneema disc.
7. **T + 60 s:** The lightened carrier drone returns to base for reload.

The 22-second climb leaves approximately 31 seconds between arrival at altitude and the predicted intercept. That interval accommodates the three-second web deployment and aerodynamic spin-up while preserving substantial timing margin.

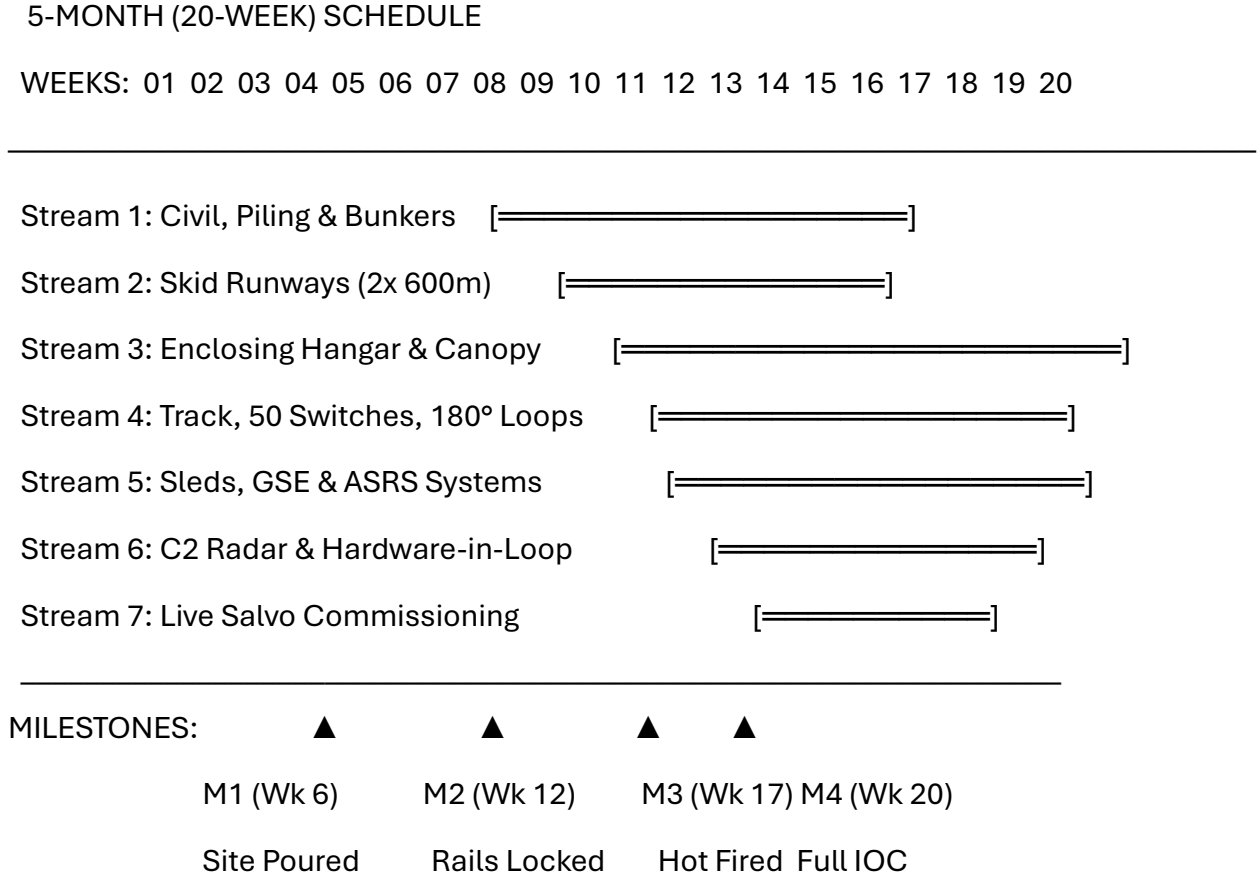
3.3.4 Conclusion

A three-second canted-fin deployment is the preferred configuration because it couples controlled unfurling with sustained autorotation. Under the stated assumptions, it reduces deployment shock and canister mass, preserves altitude during opening, and establishes a 200 m web with approximately 100–150 seconds of dwell. The proposed 0.9 mm filament and 0.70 m pitch keep the assembly near 65 kg while providing the geometric engagement needed for aerodynamic destabilization. Together, these features support a practical, persistent interception barrier with 95% kill probability against ballistic missiles.

4. Installation Schedule & Cost Breakdown (10-Track Facility)

4.1 Five-Month (20-Week) Deployment Schedule: Single 10-Track Complex

Executing the full construction and commissioning of a single 10-track fortified launch complex within a compressed 5-month (20-week) window requires aggressive parallel work streams, 24/7 multi-shift civil engineering, and pre-fabricated modular sub-assemblies.



Parallel Work-Stream Breakdown

Stream 1: Civil Engineering, Foundations & Subterranean Hardening (Weeks 1–9)

Weeks 1–3: Rapid site grading, excavation of subterranean magazine vaults, and deep-drilled caisson piling along the 10 track corridors

Weeks 4–6: Pouring high-early-strength, blast-rated reinforced concrete for the subterranean storage bunkers, launch pad grade beams, and heavy anchor blocks.

Weeks 7–9: Vault waterproofing, installation of blast-attenuating airlocks, subterranean fuel vault integration, and backfill.

Stream 2: Dual 600m Skid Runways & Thermal De-Icing (Weeks 3–10)

Weeks 3–5: Sub-base compaction and drainage grading for two parallel 600 m $\times$ 25 m recovery strips flanking the central launch site.

Weeks 6–8: Pouring high-friction concrete foundation beds and applying sacrificial polymer skid-wear friction toppings.

Weeks 9–10: Trenching and installation of ducted waste-heat plenums and continuous high-velocity Air Knife compressor manifolds along runway and track edges.

Stream 3: Enclosing Hangar Structure & Overhead Standoff Canopy (Weeks 5–15)

Weeks 5–8: Erection of outer perimeter 20 m structural steel space-frame pylons and heavy-duty ground-anchor tensioning blocks.

Weeks 9–12: Structural steel frame assembly for the full-facility enclosing hangar (covering the 10 parallel 300 m tracks, 50 switch lanes, and loop returns); roofing and side cladding installation.

Weeks 13–15: Tensioning the multi-layered high-tensile steel/Dyneema standoff net canopy across the 20 m pylons; fitting blast-baffled turbojet roof exhaust flues.

Stream 4: Track Beds, Magnetic Repulsion Rails, 50 Switches & 180° Loops (Weeks 6–14)

Weeks 6–8: Delivery and alignment of pre-fabricated steel guide rails and permanent-magnet repulsion track sections across all 10 launch lanes.

Weeks 9–11: Installation and bench-calibration of the 50 high-speed electro-mechanical track switches (5-way switchyard staging split per track).

Weeks 12–14: Erection and precision alignment of the elevated 180° banked loop-back tracks and reverse-thrust deceleration catch rails.

Stream 5: Mother Sleds, ASRS (Automated Storage and Retrieval System) Storage & GSE Outfitting (Weeks 8–16)

Weeks 8–11: Delivery, assembly, and off-track bench testing of 10 primary (plus 2 operational reserve) twin-turbojet (42 kN) reverse-thrust Mother Sled carriages.

Weeks 12–14: Installation of automated vertical drop elevators and high-density Automated Storage and Retrieval System robotic racks in the overhead mezzanine/subterranean bays.

Weeks 15–16: Fitting automated 50 kg Jet-A1 under-wing rapid-fueling hydrants, diagnostic umbilicals, and robotic ground recovery tugs.

Stream 6: C2, 3D Tracking Radar & Sensor Integration (Weeks 10–16)

Weeks 10–12: Erection of the dual-band (X/Ka) 3D active phased-array tracking radar tower and hardened optical theodolite tracking mast.

Weeks 13–14: Running redundant EMP-shielded fiber-optic fire-control buses between the central C2 command bunker and the 10 track control nodes.

Weeks 15–16: Hardware-in-the-loop (HIL) simulation tying early-warning radar feeds directly to track fire sequencing and net deployment timers.

Stream 7: System Integration, Hot-Fire Trials & IOC (Weeks 17–20)

Week 17 (Milestone 3 — Sled Hot-Fire): Static and dynamic sled acceleration runs ($a = 100$ m/s^2) on all 10 tracks; validation of clamshell reverse-thrust braking and 180° loop resets.

Week 18: Dead-load dummy separation trials at $v_0 = 245$ m/s (882 km/h); automated 5-way switch staging cycle testing under 3.0 s ripple conditions.

Week 19: Live-flight interceptor launches with empty/inert net canisters, validating rapid zoom climbs to 2 km AGL (10--14 s) and glideslope skid landings on the 600 m runways.

Week 20 (Milestone 4 — Full IOC): Live full-salvo deployment of 30 overlapping 200 m Dyneema webs at 2 km altitude; handover to operational air defense command.

Critical Path and Risk Mitigation

Critical Path Item	Major Risk	Fast-Track Mitigation Strategy
50-Switch Track Bed Alignment	Precision mechanical delays during track setting.	Off-site factory pre-assembly and laser calibration of 5-way switch modules on monolithic steel pallets prior to delivery.
Foundation Curing Time	Concrete cure delays under rail blocks.	Use self-consolidating, high-early-strength chemical additives yielding 85% design strength within 72 hours.
42 kN Sled Engine Delivery	Turbojet supply chain bottlenecks.	Parallel procurement from multiple certified mini-turbine suppliers (e.g., PBS, Motor Sich, commercial micro-turbine lines).
Radar & C2 Link Calibration	Latency jitter in sub-second fire sequencing.	Emplace standardized containerized C2 shelters running pre-compiled fire-control firmware tested extensively in simulation.

4.2 Ordnance Procurement with Dual-Warhead Mix

Daughter Interceptor Fleet (1,800 Total Units):

900 Net-Carrier Drones: $900 \times \$30{,}000 = \$27{,}000{,}000$

900 Kinetic-Warhead Drones: $900 \times \$30{,}000 = \$27{,}000{,}000$

Facility Point Defense Allocation included in total: 200 dedicated units (100 Net-Carriers / 100 Kinetic-Warhead Drones) held in local launch loops for site defense.

Total Initial Program Financial Commitment: $206,000,000 ($146M Infrastructure + $54M Ordnance).

5. Concept of Operations Against Threat Vectors: Dual-Warhead Optimization

The C2 system dynamically assigns either Net-Carrier or Kinetic-Warhead interceptors based on threat geometry, target hardening, collateral risk, and target speed.

5.1 One-Way Attack (OWA) Drones and Loitering Munitions

Optimal Weapon Selection: Net-Carrier Interceptor (Primary) / Kinetic-Warhead Interceptor (Urban Proximity).

Tactical Approach: For slow, low-RCS targets (e.g., Shahed-136) over open territory, net-carriers deploy high-tensile Dyneema barrier screens, entangling prop-rotors without explosive blast hazards. When targets approach populated urban zones or critical infrastructure where unexploded drone warheads pose secondary risks, kinetic-warhead interceptors execute direct-impact strikes to detonate the target payload in mid-air.

5.2 Cruise Missiles

Optimal Weapon Selection: Kinetic-Warhead Interceptor (Primary) / Net-Carrier Interceptor (Narrow Chokepoints).

Tactical Approach: Subsonic/supersonic low-altitude terrain-following cruise missiles require high-energy structural destruction. Kinetic-warhead interceptors use proximity-fuzed blast-fragmentation to sever guidance systems and warhead buses. In narrow terrain corridors or mountain passes, net-carriers lay dense barrier screens to shear flight actuators and air intakes.

5.3 Glide Bombs (e.g., KAB/FAB with UMPK Kits)

Optimal Weapon Selection: Net-Carrier Interceptor (High Altitude) / Kinetic-Warhead Interceptor (Heavy Penetrators).

Tactical Approach: At high release altitudes, net-carriers deploy vertical Dyneema screens directly across the glide path, shearing off high-aspect-ratio control wings and causing the bomb to tumble short of its target. Against heavy armored penetrator casings (FAB-1500/3000), kinetic-warhead interceptors deliver high-density kinetic impacts to fracture control surfaces or force pre-detonation.

5.4 Ballistic Missiles

Optimal Weapon Selection: Net swarm.

Tactical Approach: Ballistic intercepts utilize a 30-net array spread over a 1 km diameter, as described in Section 3.3 above. OJEM is the most favorable cost-exchange system proposed for ballistic-missile defense. As such, timely development and validation should be an urgent priority for defensive systems worldwide.

6. Concept of Operations: Frontline Deployment with 50 km Standback

When deployed 50 km behind the Forward Line of Own Troops (FLOT), the 10-track array functions as a dual defensive umbrella and responsive strike hub:

Survivability & Dispersal: Positioned 50 km behind the FLOT, the facility remains beyond standard tube artillery (30 km). Incoming rocket artillery or loitering attacks are countered by the 20 m-draped Dyneema canopy net and the 200 dedicated point-defense interceptors (100 net / 100 kinetic).

Paired Salvo Employment: Operating with a 150 km $-$ 250 km combat radius, the facility launches dual-ship elements (1 Net / 1 Kinetic) at a 3-second per-track cadence (200 launches/minute array-wide), providing simultaneous soft-kill area suppression and hard-kill precision targeting over 100 km of front line.

7. Offensive Concept of Operations

The 50/50 warhead split enables seamless transitioning between offensive defense-suppression and direct precision strike missions.

7.1 Tactical Frontline Offense

Kinetic Precision Strikes: Kinetic-warhead interceptors execute direct, high-speed dive attacks against enemy air-defense radar nodes, command posts, artillery positions, and armored concentrations within a 200 km strike radius.

Defensive Net Suppression: Net-carrier interceptors fly lead positions, deploying barrier screens over enemy launch sites to neutralize defending counter-drone swarms, reconnaissance UAVs, and attack helicopters before friendly ground forces advance.

7.2 Long-Range Theater Attack Escort

Mixed Penetration Corridors: In deep theater strike missions, mixed waves of interceptors precede friendly cruise missiles or bomber formations. Net-carrier drones clear defending surface-to-air missile zones and airborne threats, while kinetic-warhead drones eliminate enemy Airborne Early Warning (AEW) aircraft and key radar installations.

8. Summary of Overall Impact across Conflict Theaters

Across all theaters, the interceptor's low unit cost and return-to-base capability produce a cost-exchange ratio that strongly favors the defender, addressing a central vulnerability of current defensive systems. The OJEM system also provides a defense against ballistic-missile attacks, blunting the effect of the most consequential threats in contemporary conflict theaters.

8.1 Ukraine Operational Theater



Impact: Provides an economical solution to the magazine-depth crisis. Net-carriers neutralize massed Shahed-136 UAVs without wasting explosive warheads, preserving kinetic-warhead interceptors to destroy high-speed UMPK glide bombs and tactical

ballistic missiles. This dual capability protects critical infrastructure while conserving high-cost long-range SAMs (e.g., Patriot, NASAMS).

The Economic & Strategic Logic: From stalemate to dominance

Waiting out Moscow through asymmetric fire dominance relies on three compounding pressures:

THE TRIPARTITE CONTAINMENT VISE

[1. Frontline Drain] —► Constant material attrition ($8,500 sortie vs. T-90/BM-21)

[2. Economic Overhead] —► Sustaining millions of tons of supply under drone siege

[3. Sanctions & Stagnation] ► $300B+ sovereign freeze + demographic/capital exhaustion

The Cost-Exchange Deficit: When Ukraine can destroy a $3M infantry fighting vehicle or a $500k radar system with an $8,500 sled-launched drone, the financial burden of occupation grows exponentially for the attacker.

The "East Germany" Precedent: The Soviet Union did not relinquish Eastern Europe or the Baltic states because NATO tanks smashed through the Fulda Gap; it surrendered them because the economic and military cost of maintaining the occupation exceeded the political survival capacity of the regime.

Preserving Ukrainian Human Capital: Avoiding suicidal, high-casualty frontal assaults against concrete minefields preserves Ukraine’s working-age demographic, allowing the nation to rebuild its industrial, agricultural, and tech base west of the defensive line.

Key Risks to Manage in an Extended "Containment" Strategy

While this strategy is historically proven, maintaining a 5-to-15-year containment posture requires mitigating specific long-term liabilities:

Risk Domain	Long-Term Threat	Strategic Countermeasure
Western Fatigue	Political shifts in donor nations reducing aid tranches.	Transition drone and sled production to 100% domestic/subterranean manufacturing.
Demographic Attrition	Prolonged low-intensity conflict suppressing birth rates and investment.	Full air-defense insulation (canted-fin net barriers + Patriot) over Kyiv/Lviv to normalize civilian life.
Adversary Adaptation	Russia heavily automating its own supply chains with underground UGVs.	Scale fiber-optic forward nodes to identify and seal subterranean tunnel exits.
Border Normalization	World powers de facto accepting the occupied line as permanent.	Maintain non-recognition treaties and keep economic sanctions strictly tied to territorial integrity.

Summary

Deploying a multi-site active containment grid takes the initiative away from Russian mass.

It turns the occupied territories from a Russian forward staging base into a bleeding liability—securing 80%+ of free Ukraine as a thriving, technologically fortified European state while setting the mechanical conditions for diplomatic restoration when the political center in Moscow eventually fractures under the weight of sustained attrition.



8.2 Gulf War/Middle East Operational Theater

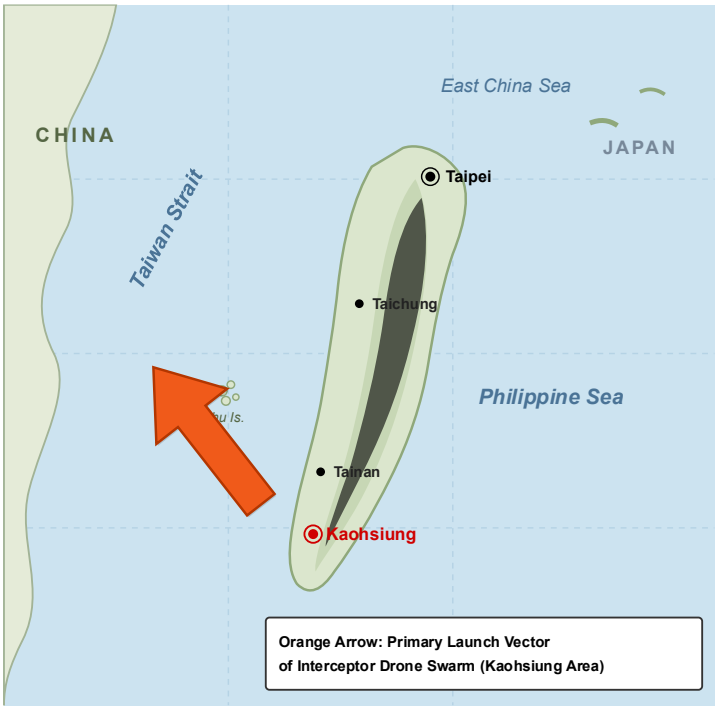
Impact: Secures vital maritime chokepoints (Strait of Hormuz, Bab al-Mandab) and energy infrastructure. Net-carriers neutralize dense adversary loitering drone swarms over commercial shipping lanes with zero explosive fallout, while kinetic-warhead interceptors destroy incoming anti-ship cruise missiles before they reach task force radar horizons. Gulf nations can develop cross land Strait bypass systems under an invincible defensive umbrella protecting both construction and operation.

8.3 Taiwan Conflict Theater

Use Case 3:

Taiwan Defense: Coastal OJEM Array Launch Vectors against Sea Invasion

- An integrated ground-based array provides a dense, scalable defense.
- The orange arrow indicates the primary launch direction of defensive interceptor drone swarms from a fortified coastal OJEM array in southern Taiwan.
- The system is designed to intercept and neutralize large-scale incoming saturation threats (ships, missiles, drone swarms) in the Taiwan Strait before they approach the coastline.
- Arrays are strategically positioned along the western and northern coasts to establish overlapping defense screens.



Impact: Counters China's anti-access/area-denial (A2/AD) missile doctrine. Positioned along Taiwan's mountain corridors and coastal sectors, the 10-track system deploys kinetic-warhead interceptors to destroy primary land-attack cruise missiles, while net-carriers neutralize secondary drone swarms and air-dropped munitions, ensuring defensive dominance against cross-strait amphibious operations.

9. Capital Expenditure (CapEx)

A. Heavy Civil Engineering ($55M)

Subterranean Blast Bunkers: Housing a 300-drone ready magazine, fuel tank farms, and crew quarters beneath reinforced Class 1/Div 1 blast-rated concrete covers.

B. Overhead Kinetic Standoff Net Superstructure ($25M)

Covering a multi-hectare industrial site (a $300+$ m track footprint plus 600 m recovery corridors) at a height of 200 meters requires pylons, perimeter ground anchorages, and thousands of square meters of multi-layered steel/Dyneema standoff mesh.

C. Automated Switchyard & 180° Loop Mechanics ($18M)

A 5-way split across 10 tracks equals 50 high-speed, precision-aligned track switch mechanisms with automated linear actuators, interlocks, and overhead banked loop structures engineered for 30 m/s dynamic sled entry.

D. Automated Factory GSE & High-Speed Fueling Hydrants ($20M)

Explosion-proof overhead gantry cranes for rapid 15-second interceptor drops, automated 50 kgJet-A1 high-pressure under-wing fueling couplers, and robotic drone recovery tugs.

E. Local Sensor, Radar & Fire-Control Integration ($15M)

Dedicated dual-band (X/Ka) 3D active phased-array tracking radar, high-speed optical theodolite masts, and fiber-optic fire-control links to integrate with national early-warning radar arrays (e.g., Patriot/SAMP-T/NATO data feeds).

9.1 Capital Expenditure Summary

CapEx accounts for the airframe procurement, proportional magazine expansion, and the enlarged subterranean civil engineering footprint required to store, fuel, and service 1,800 airframes per complex under full standoff protection.

Subsystem / Cost Category	Per Site Cost (1,800 Drones)	10-Site Nationwide Program (18,000 Drones)	Scope & Technical Deliverables
Subterranean Bunkering & Civil Hardening	$80.0M	$800.0M	Deep blast caissons, high-density underground ASRS vault expansion, reinforced grade beams.
Overhead Standoff Canopy Superstructure	$25.0M	$250.0M	20 m high-strength steel pylons, tension anchors, full-facility kinetic/drone mesh.
Launch Track Array (10 Tracks, 50 Switches, Loops)	$30.0M	$300.0M	10 × 300 m permanent-magnet rails, 5-way staging switchyard, 180° loop-back rails.
2 × 600 m Skid Runways & Air-Knife De-Icing	$15.0M	$150.0M	Sacrificial polymer tarmac, ducted turbojet waste-heat plenums, high-pressure air knives.
Hangar Automation, High-Density ASRS & GSE	$35.0M	$350.0M	Automated vertical drop loaders, robotic ASRS retrieval racks, fuel manifolds, recovery tugs.
Local 3D Tracking Radar & Hardened C2 Node	$15.0M	$150.0M	Dual-band (X/Ka) millimeter-wave radar mast, optical tracking gate, EMP-shielded C2 bunker.
Interceptor Fleet (1,800 drones/site @ $35k)	$63.0M	$630.0M	18,000 total reusable airframes (1.6 kN turbojet, composite belly skids, avionics).
Mother Sleds (10 sleds/site @ $150k)	$1.5M	$15.0M	100 total twin-turbojet (42 kN) reverse-thrust sleds with Halbach magnetic guide arms.
Strategic Net Magazine (9,000 units/site @ $6.5k)	$58.5M	$585.0M	90,000 total vacuum canisters (65 kg Dyneema SK99 net, canted fins, sabots).
System Integration, NRE, High-Volume Tooling	$30.0M	$300.0M	High-rate manufacturing tooling, test benches, 20% critical line spares, site commissioning.
TOTAL CapEx	**$353.0M**	**$3,530.0M ($3.53 Billion)**	**Turnkey, fully hardened 18,000-interceptor nationwide defense network.**

10. Operating Expenditure (OpEx)

10.1 Annual Standing Readiness and Maintenance

With 18,000 airframes and 90,000 sealed net canisters distributed across 10 high-density automated sites, recurring operational overhead increases to accommodate scheduled turbine inspections, robotic ASRS cycling, and facility operations.

Base Facility & Track Infrastructure Upkeep: $35.0M / year ($3.5M per site for track maintenance, rail alignment, and heating/air-knife systems).

Airframe Fleet Maintenance & Scheduled Overhauls: $54.0M / year (~$3,000 per airframe annually for turbine bearing replacements, avionics calibration, and skid refurbishments).

Magazine Environmental Management & Desiccant Cycling: $9.0M / year (Nitrogen purge monitoring and vacuum seal verification across 90,000 canisters).

Crew, C2 Mesh Telemetry, Power & Base Security: $22.0M / year ($2.2M per site).

Total Annual Standing OpEx: $120.0 Million / year ($12.0M per site annually).

10.2 Marginal Engagement Economics (per 30-Net Salvo)

Because the airframe is reusable and the ground launch system is powered by short-duration jet fuel, the marginal cost per engagement does not change with fleet size:

$$\text{Cost}_{\text{salvo}} = 30 \times \$6{,}500 \text{ (Nets)} + 30 \times \$18 \text{ (Sled Fuel)} + 30 \times \$75 \text{ (Drone Fuel)} + \text{Airframe Depr. } (30 \times \$3{,}500) = \$\mathbf{302{,}790}$$

Cost to Defeat a 2-Missile Ballistic Salvo: ~$303,000

Threat Asset Value Neutralized (2x Iskander-M / Kinzhal): ~$7.0M to $9.0M

Cost-Exchange Ratio: $> 23{:}\,1$ in favor of the defense

Strategic Impact of the 18,000-Unit Expansion

Immunity to Saturation Depletion: A 90,000-net magazine allows Ukraine to fire 3,000 full 30-net defensive salvos. At Russia's maximum production rate of ~120 ballistic missiles per month, this inventory provides over 4 years of continuous saturation defense without requiring industrial resupply.

Multi-Wave Salvo Endurance: With 1,800 airframes per site, a single complex can absorb consecutive, multi-wave saturation raids (launching up to 60 airframes per minute across

its 10 tracks) for hours without waiting for returned airframes to complete turnaround cycles.

Strategic Conclusion: OJEM combines a reusable launch architecture, deep interceptor and net magazines, persistent area coverage, and low marginal engagement cost. The projected economics—approximately $303,000 to counter a two-missile salvo valued at $7–9 million—would reverse the prevailing cost equation by allowing the defender to sustain repeated engagements without spending more per salvo than the attacker. This combination represents a game-changing advance: practical ballistic-missile defense with the magazine depth and favorable cost-exchange ratio required for sustained operations.

Attachment A — Acronym List

This attachment expands acronyms used in the document, including terms defined in the body and terms verified against authoritative aviation, defense, and technical glossaries.

Acronym	Meaning
A2/AD	Anti-access/area denial
AEW	Airborne Early Warning
AGL	Above Ground Level
ASRS	Automated Storage and Retrieval System
C2	Command and Control
CapEx	Capital Expenditure
EMP	Electromagnetic Pulse
FLOT	Forward Line of Own Troops
GSE	Ground Support Equipment
HIL	Hardware-in-the-Loop
IOC	Initial Operational Capability
NATO	North Atlantic Treaty Organization
NRE	Non-Recurring Engineering
OJEM	Open-air JEt Mother/daughter
OpEx	Operating Expenditure
OWA	One-Way Attack
PDE	Pulse Detonation Engine
PM	Permanent Magnet
RCS	Radar Cross Section
RTB	Return to Base
SAM	Surface-to-Air Missile
SAMP-T	Surface-to-Air Missile Platform/Terrain
UAV	Uncrewed Aerial Vehicle
UGV	Uncrewed Ground Vehicle
UMPK	Unified Gliding and Correction Module

Verification sources: NATO Glossary of Abbreviations; FAA Definitions; DoD Integrated Program Management Acronyms; and Automated Storage and Retrieval Systems.

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